Lab:15.1 Name :V.Pranavith

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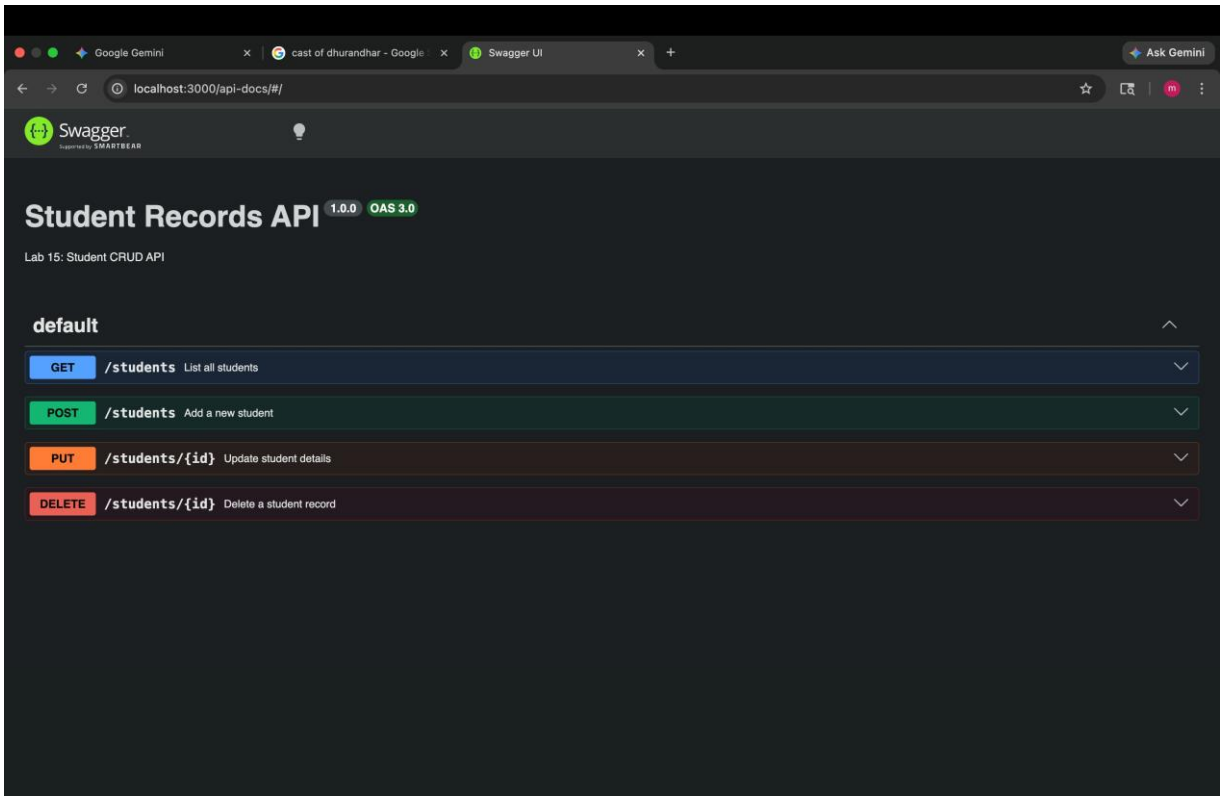
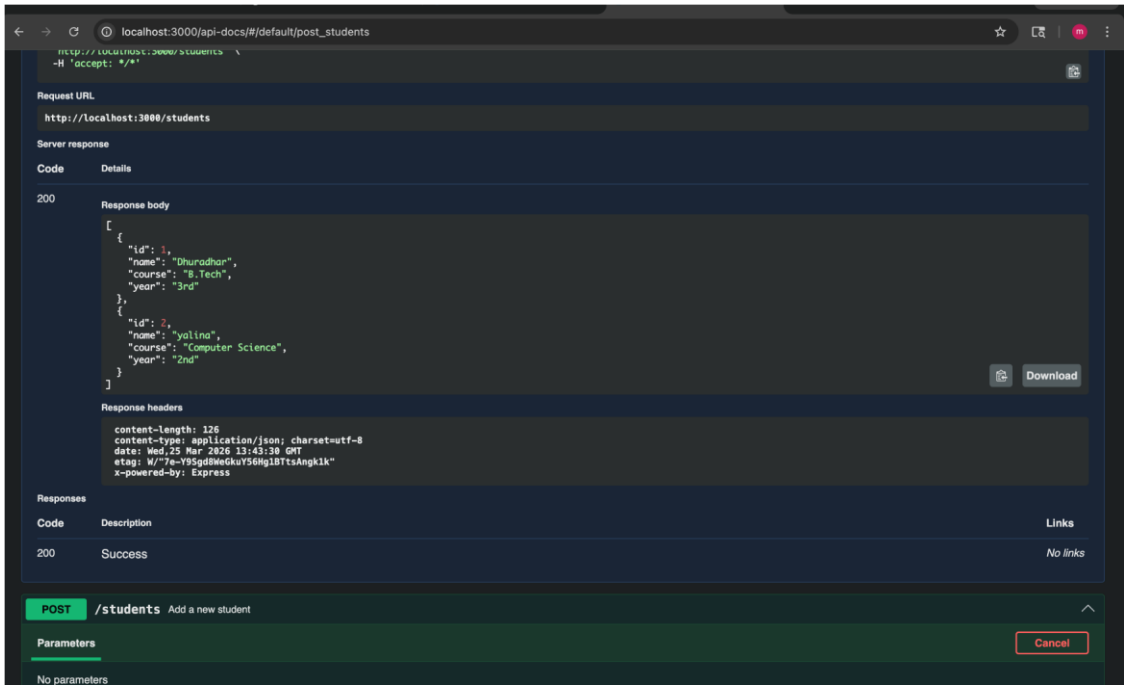
Backend API Development: Creating RESTful services with AI

Task1: Student Records API





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localhost:3000/api-docs/#/default/post_students

Parameters | Cancel | Reset

No parameters

Request body required — application/json

Edit Value | Schema

```
{
  "name": "Disha",
  "course": "B-Tech",
  "year": "3rd year"
}
```

Execute

Responses

Code	Description	Links
201	Created	No links

PUT /students/{id} Update student details

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localhost:3000/api-docs/#/

Request URL

http://localhost:3000/students

Server response

Code	Details
200	

Response body


```
[
  {
    "id": 1,
    "name": "Dhuradhar",
    "course": "B.Tech",
    "year": "3rd"
  },
  {
    "id": 2,
    "name": "yalina",
    "course": "Computer Science",
    "year": "2nd"
  },
  {
    "id": 3,
    "name": "Disha",
    "course": "B-Tech",
    "year": "3rd year"
  },
  {
    "id": 4,
    "name": "Disha",
    "course": "B-Tech",
    "year": "3rd year"
```


Download

Response headers

```
connection: keep-alive
content-length: 246
content-type: application/json; charset=utf-8
date: Wed,25 Mar 2026 13:45:27 GMT
etag: W/"f6-tV/081jyf0V9eEiM/tBhBeFiirk"
keep-alive: timeout=5
x-powered-by: Express
```

Responses

Code	Description	Links
200	Success	No links

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localhost:3000/api-docs/#/default/put_students__id_

Responses

Curl

```
curl -X 'PUT' \
  'http://localhost:3000/students/3' \
  -H 'accept: */*'
```

Request URL

http://localhost:3000/students/3

Server response

Code	Details
200	Response body

```
{
  "id": 3,
  "name": "Disha",
  "course": "B-Tech",
  "year": "3rd year"
}
```

Download

Response headers

```
connection: keep-alive
content-length: 59
content-type: application/json; charset=utf-8
date: Wed,25 Mar 2026 13:46:18 GMT
etag: W/"3b-8kr86eI4UcSinxn5NsGAoXfibGM"
keep-alive: timeout=5
x-powered-by: Express
```

Responses

Code	Description	Links
200	Updated	*No links*

DELETE /students/{id} Delete a student record

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localhost:3000/api-docs/#/default/delete_students__id_

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:3000/students' \
  -H 'accept: */*'
```

Request URL

http://localhost:3000/students

Server response

Code	Details
200	Response body

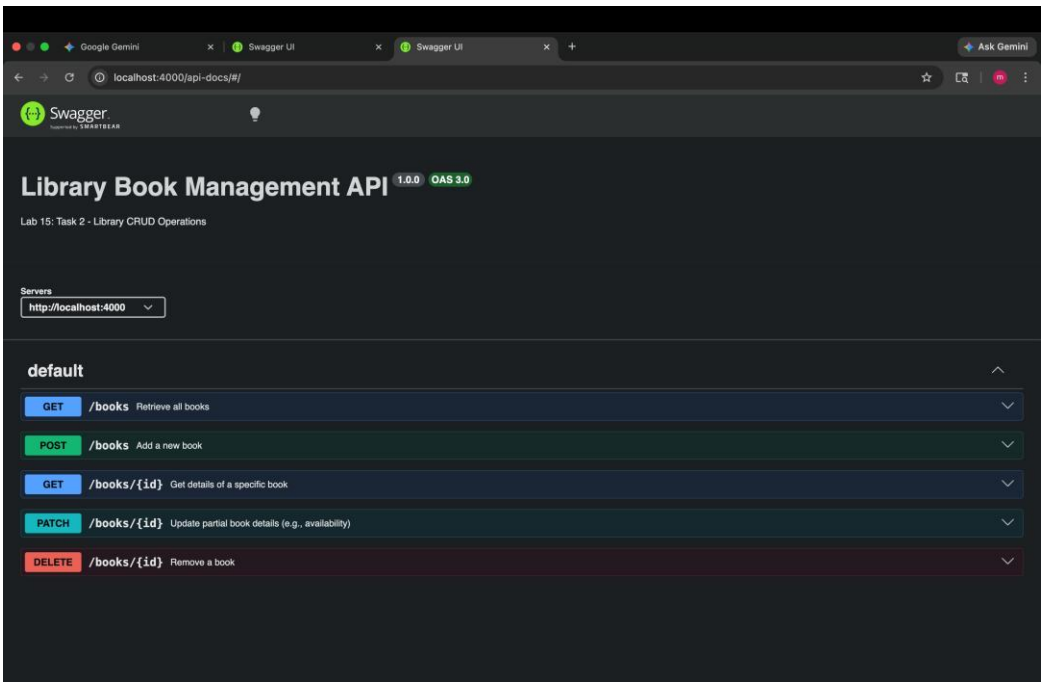
```
[
  {
    "id": 1,
    "name": "Dhuradhar",
    "course": "B.Tech",
    "year": "3rd"
  },
  {
    "id": 2,
    "name": "yalina",
    "course": "Computer Science",
    "year": "2nd"
  },
  {
    "id": 4,
    "name": "Disha",
    "course": "B-Tech",
    "year": "3rd year"
  }
]
```

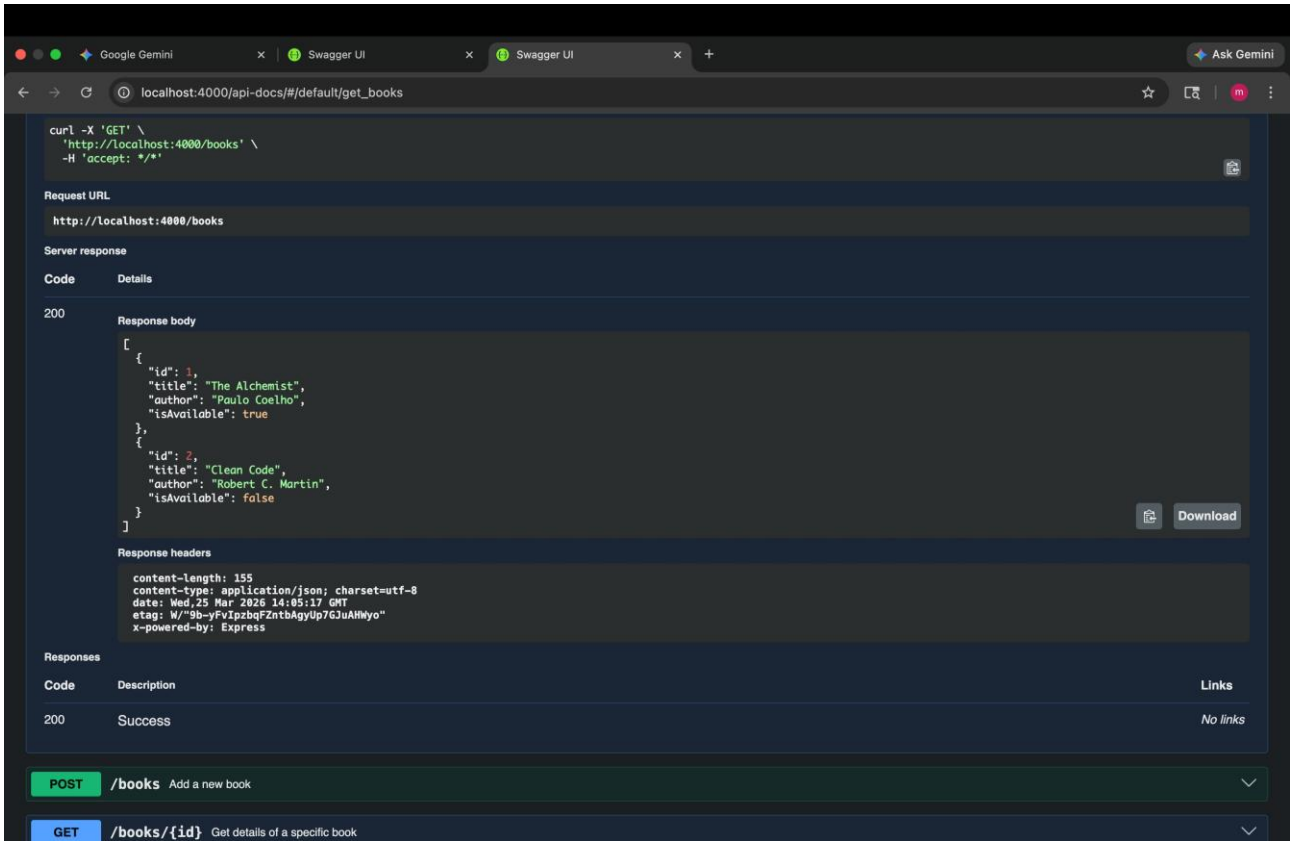
Download

Response headers

```
connection: keep-alive
content-length: 186
content-type: application/json; charset=utf-8
date: Wed,25 Mar 2026 13:46:57 GMT
etag: W/"ba-kHy0HNtmjEcVvRIMQknskBXAfzY"
keep-alive: timeout=5
x-powered-by: Express
```

Task 2:Library Book Management API





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localhost:4000/api-docs/#/default/post_books

POST /books Add a new book

Parameters | Cancel | Reset

No parameters

Request body required | application/json

Edit Value | Schema

```
{
  "title": "DSA",
  "author": "TEJAS"
}
```

Execute | Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:4000/books' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
```

Google Gemini | Swagger UI | Swagger UI | Ask Gemini

localhost:4000/api-docs/#/default/get_books__id_

GET /books/{id} Get details of a specific book

Parameters | Cancel

Name	Description
id * required integer *(path)*	4

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:4000/books/4' \
  -H 'accept: */*'
```

Request URL

http://localhost:4000/books/4

Server response

Code	Details
200	Response body

```
{
  "id": 4,
  "title": "DSA",
  "author": "TEJAS",
  "isAvailable": true
}
```

Download

Response headers

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localhost:4000/api-docs/#/default/patch_books__id_

Responses

Curl

```
curl -X 'PATCH' \
  'http://localhost:4000/books/1' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
  "isAvailable": true
}'
```

Request URL

http://localhost:4000/books/1

Server response

Code	Details
200	Response body

```
{
  "id": 1,
  "title": "The Alchemist",
  "author": "Paulo Coelho",
  "isAvailable": true
}
```

Download

Response headers

```
connection: keep-alive
content-length: 75
content-type: application/json; charset=utf-8
date: Wed,25 Mar 2026 14:08:41 GMT
etag: W/"4b-91/0hMGNr0Sa5RETcHC8yCLNq3w"
keep-alive: timeout=5
x-powered-by: Express
```

Responses

Code	Description	Links
200	Updated	*No links*

DELETE /books/{id} Remove a book

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localhost:4000/api-docs/#/default/delete_books__id_

Parameters Cancel

Name	Description
id * required integer *(path)*	3

Execute | Clear

Responses

Curl

```
curl -X 'DELETE' \
  'http://localhost:4000/books/3' \
  -H 'accept: */*'
```

Request URL

http://localhost:4000/books/3

Server response

Code	Details
200	Response body

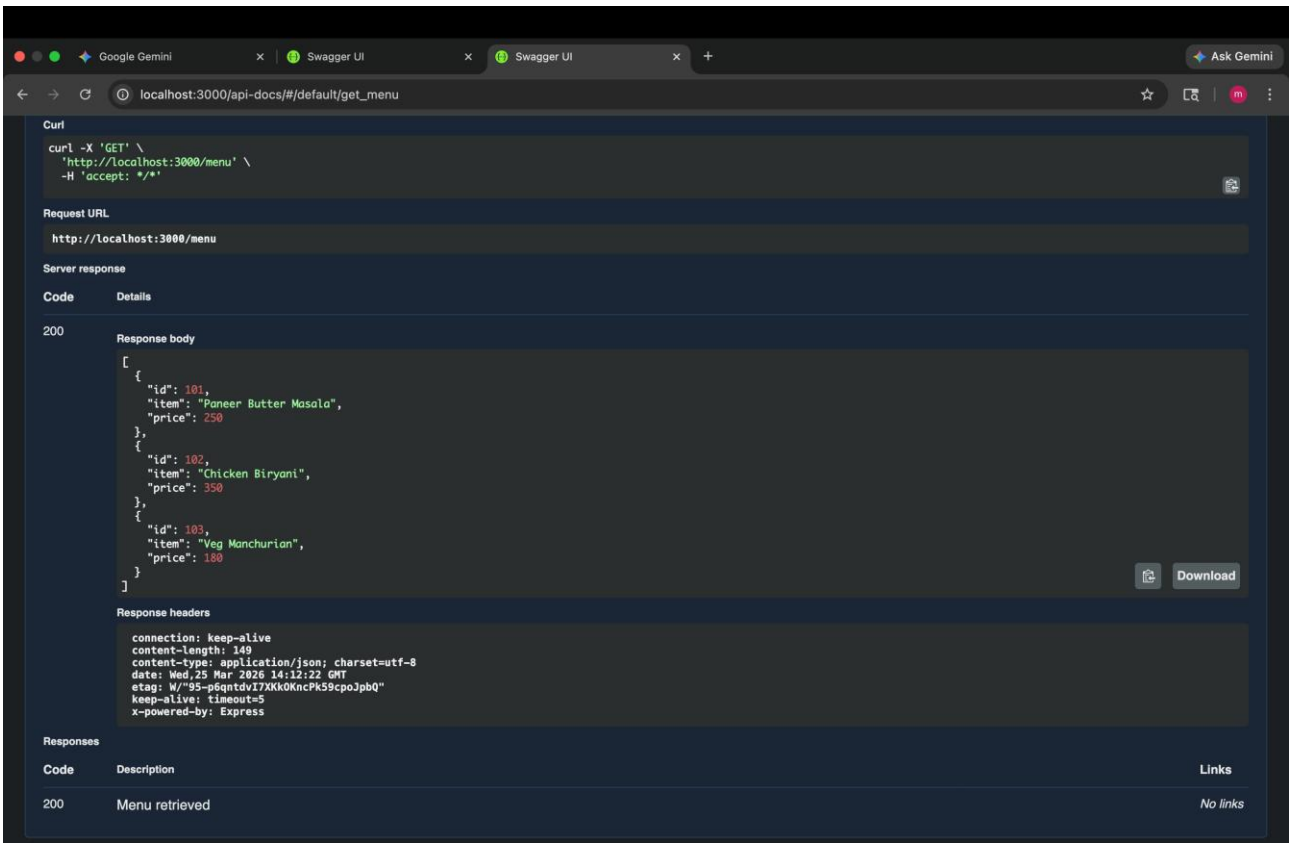
```
{
  "message": "Book deleted"
}
```

Download

Response headers

```
connection: keep-alive
content-length: 26
content-type: application/json; charset=utf-8
date: Wed,25 Mar 2026 14:09:39 GMT
etag: W/"1a-ASXGQesuMor5Uhx4P51EVemPQAU"
keep-alive: timeout=5
```

Task 4 – Real-Time Application: Online Food Ordering API



Google Gemini | Swagger UI | Swagger UI | Ask Gemini

localhost:3000/api-docs/#/default/post_order

```
  -H 'Content-Type: application/json' \
  -d '{
  "itemIds": [
    2
  ]
}'
```

Request URL

http://localhost:3000/order

Server response

Code	Details
201	**Response body**

```
{
  "id": 2,
  "items": [],
  "status": "Preparing",
  "orderTime": "2026-03-25T14:13:32.448Z"
}
```

Download

Response headers

```
connection: keep-alive
content-length: 79
content-type: application/json; charset=utf-8
date: Wed,25 Mar 2026 14:13:32 GMT
etag: W/"4f-YElh6V8689vOz+fFyoxFLCXgImY"
keep-alive: timeout=5
x-powered-by: Express
```

Responses

Code	Description	Links
201	Order Placed	*No links*

GET /order/{id} Track order status

PUT /order/{id} Update an existing order

DELETE /order/{id} Cancel an order

Google Gemini | Swagger UI | Swagger UI | Ask Gemini

localhost:3000/api-docs/#/default/get_order__id_

Parameters | Cancel

Name	Description
id * required integer (path)	1

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:3000/order/1' \
  -H 'accept: */*'
```

Request URL

```
http://localhost:3000/order/1
```

Server response

Code	Details
200	Response body

```
{
  "id": 1,
  "status": "Preparing"
}
```

Download

Response headers

```
content-length: 29
content-type: application/json; charset=utf-8
date: Wed,25 Mar 2026 14:14:18 GMT
etag: W/"1d-kgH/iu1wwxpB6tRUP1r7HJ/ilPI"
x-powered-by: Express
```

Responses

Google Gemini | Swagger UI | Swagger UI | Ask Gemini

localhost:3000/api-docs/#/default/put_order__id_

Execute | Clear

Responses

Curl

```
curl -X 'PUT' \
  'http://localhost:3000/order/1' \
  -H 'accept: */*'
```

Request URL

```
http://localhost:3000/order/1
```

Server response

Code	Details
200	Response body

```
{
  "id": 1,
  "items": [],
  "status": "Updated/Modified",
  "orderTime": "2026-03-25T14:13:30.994Z"
}
```

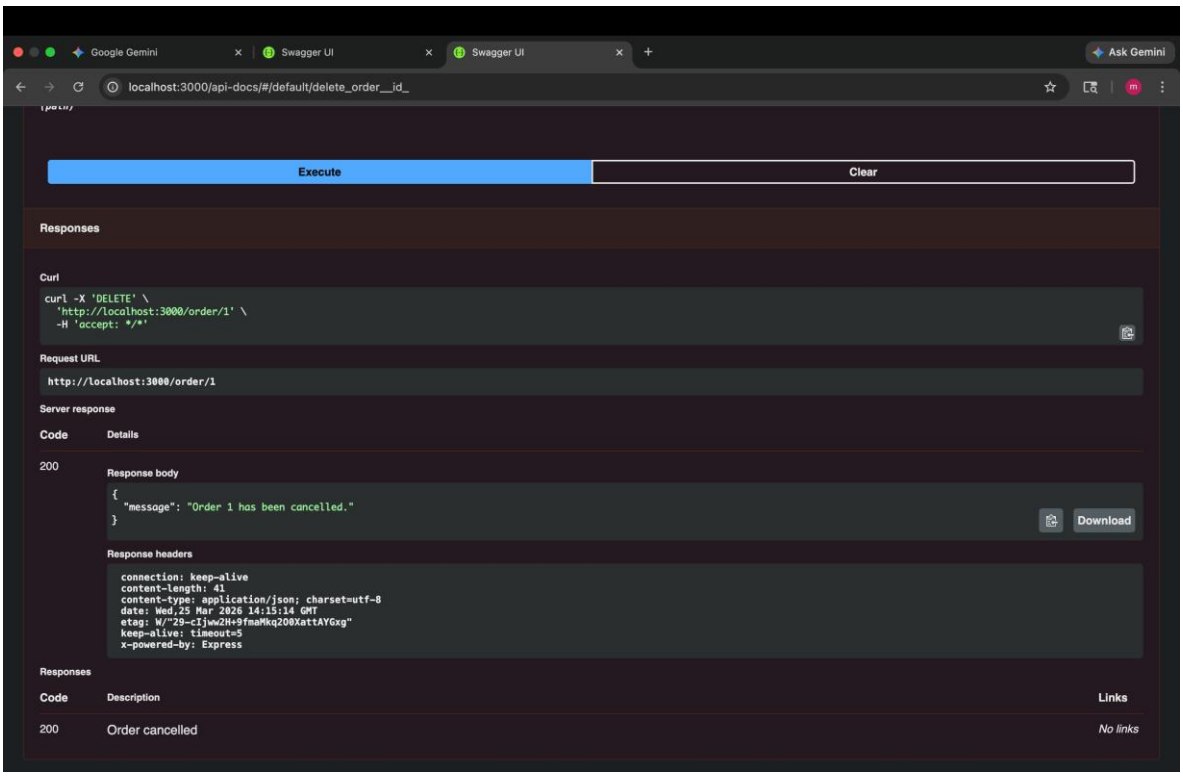
Download

Response headers

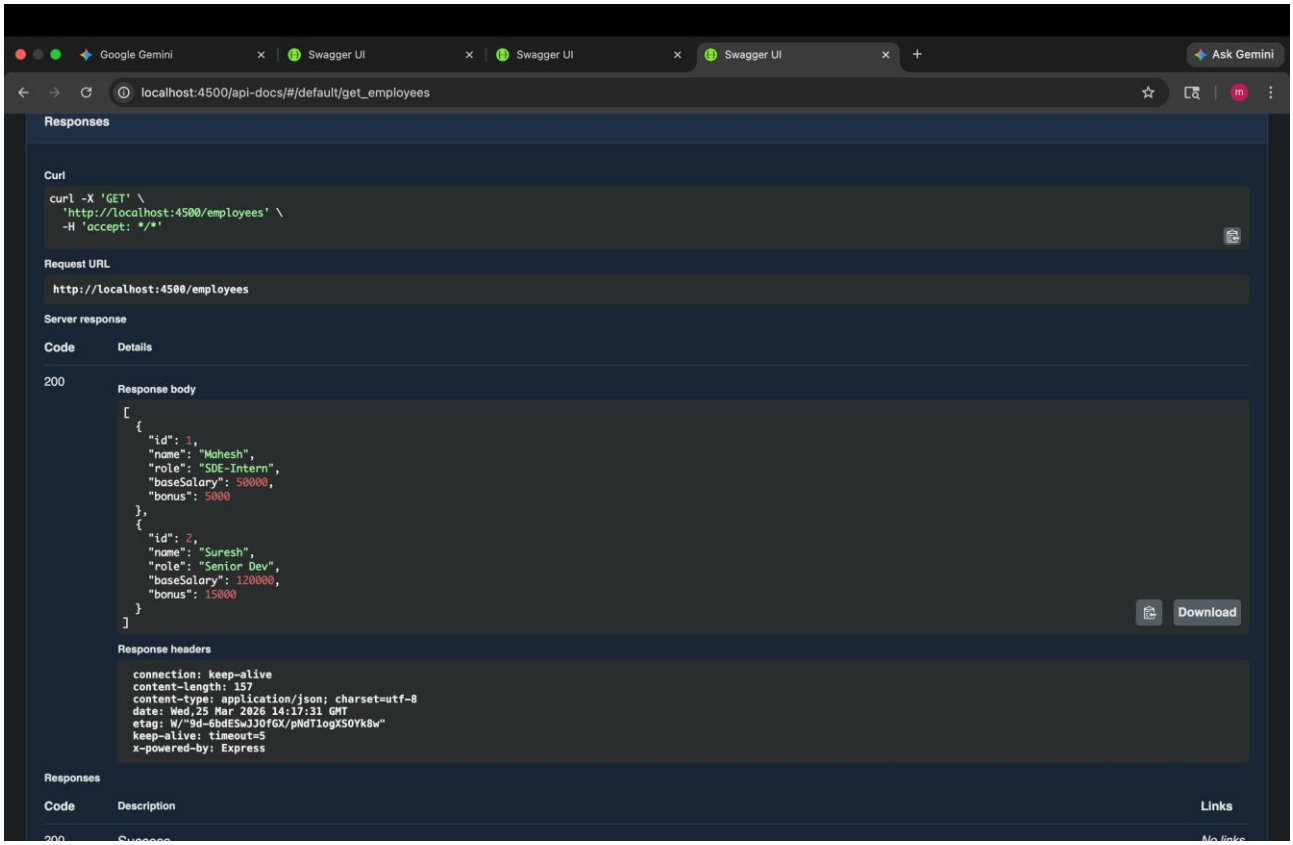
```
connection: keep-alive
content-length: 86
content-type: application/json; charset=utf-8
date: Wed,25 Mar 2026 14:14:43 GMT
etag: W/"56-vI07w/GW30bVKmbfY31rN0qMR48"
keep-alive: timeout=5
x-powered-by: Express
```

Responses

Code	Description	Links
200	Order updated	*No links*



Task3: Employee Payroll API



Google Gemini | Swagger UI | Swagger UI | Swagger UI | Ask Gemini

localhost:4500/api-docs/#/default/post_employees

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:4500/employees' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": "Naresh",
  "role": "HR",
  "baseSalary": 120000,
  "bonus": 10000
}'
```

Request URL

http://localhost:4500/employees

Server response

Code	Details
201	Response body

```
{
  "id": 3,
  "name": "Naresh",
  "role": "HR",
  "baseSalary": 120000,
  "bonus": 10000
}
```

Download

Response headers

```
connection: keep-alive
content-length: 70
content-type: application/json; charset=utf-8
date: Wed,25 Mar 2026 14:20:22 GMT
etag: W/"46-DNO8tS3kh0cfcenhA4tzQ8UjKGk"
keep-alive: timeout=5
x-powered-by: Express
```

Responses

Code	Description	Links
201	Employee Created	*No links*

Google Gemini | Swagger UI | Swagger UI | Swagger UI | Ask Gemini

localhost:4500/api-docs/#/default/put_employees__id__salary

Responses

Curl

```
curl -X 'PUT' \
  'http://localhost:4500/employees/1/salary' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
  "baseSalary": 5000000,
  "bonus": 100000
}'
```

Request URL

http://localhost:4500/employees/1/salary

Server response

Code	Details
200	Response body

```
{
  "message": "Salary updated successfully",
  "employee": {
    "id": 1,
    "name": "Mahesh",
    "role": "SDE-Intern",
    "baseSalary": 5000000,
    "bonus": 100000
  }
}
```

Download

Response headers

```
connection: keep-alive
content-length: 133
content-type: application/json; charset=utf-8
date: Wed,25 Mar 2026 14:21:09 GMT
etag: W/"85-qhWACs6ohaO5S6Q0S+izuYcILfo"
keep-alive: timeout=5
x-powered-by: Express
```

Responses

Code	Description	Links

Google Gemini | Swagger UI | Swagger UI | Swagger UI | Ask Gemini

localhost:4500/api-docs/#/default/delete_employees__id_

(path)

Execute | Clear

Responses

Curl

```
curl -X 'DELETE' \
  'http://localhost:4500/employees/2' \
  -H 'accept: */*'
```

Request URL

```
http://localhost:4500/employees/2
```

Server response

Code	Details
200	

Response body

```
{
  "message": "Employee 2 removed from system."
}
```

Download

Response headers

```
connection: keep-alive
content-length: 45
content-type: application/json; charset=utf-8
date: Wed,25 Mar 2026 14:21:39 GMT
etag: W/"2d-tdjIBmOHcCUg9/wgy+6ghKRPwA8"
keep-alive: timeout=5
x-powered-by: Express
```

Responses

Code	Description	Links
200	Employee Removed	*No links*